differences in brightness or light intensity. The phase changes are not detectable to human eye whereas the brightness or light intensity can be easily detected.

2.1.2 Optical Components of PCM

The phase contrast microscope is similar to an ordinary compound microscope in its optical components. It possesses a light source, condenser system, objective lens system and ocular lens system (Figure 2.1).

A phase contrast microscope differs from bright field microscope in having,

i. Sub-stage annular diaphragm (phase condenser)

   An annular aperture in the diaphragm is placed in the focal plane of the sub-stage which controls the illumination of the object. This is located below the condenser of the microscope. This annular diaphragm helps to create a narrow, hollow cone of light to illuminate the object.

ii. Phase – plate (diffraction plate or phase retardation plate)

   This plate is located at the back focal plane of the objective lenses. The phase plate has two portions, in which one is coated with light retarding material (Magnesium fluoride) and the other portion devoid of light retarding material but can absorb light. This plate helps to reduce the phase of the incident light (Figure 2.2).

2.1.3 Working Mechanism of Phase Contrast Microscopy

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The unstained cells cannot create contrast under the normal microscope. However, when the light passes through an unstained cell, it encounters regions in the cell with different refractive indexes and thickness. When light rays pass through an area of high refractive index, it deviates from its normal path and such light rays experience phase change or phase retardation (deviation). Light rays pass through the area of less refractive index

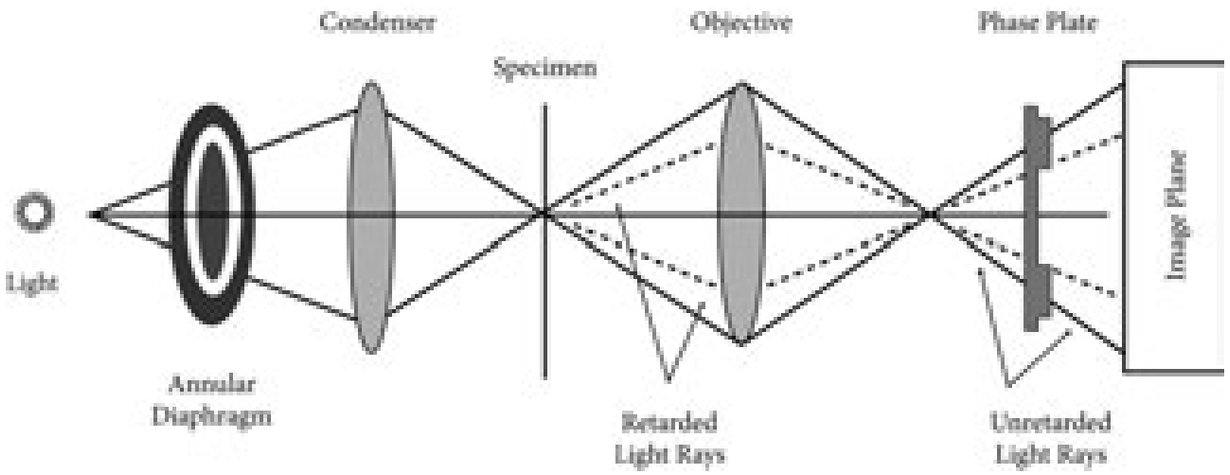


Figure 2.1: Components of phase contrast microscope

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